PSE Student Workbook – Modules 1, 2 & 3

Completed Project: Emergency Fuel Assistance for Stranded Vehicles

Super System: Transportation and Roadside Mobility Support System

Sub-System: Emergency Fuel Assistance for Stranded Vehicles

Scope: National highways and major roads in India; initial study context focused on Telangana/Hyderabad-region road travel.

Project focus: Understand the existing roadside fuel-access problem and design a resilient system response. The workbook requires the team to study the system before committing to a technology solution.

Prepared for: Problem Solving in Engineering (PSE) – IP-3 Experiential Learning Course

MODULE 1 – Project Informatics Document (PID)

1. Affected System / Problem Area Selection

Workbook field	Filled content
Super System	Transportation and Roadside Mobility Support System
Specific Sub-System	Emergency Fuel Assistance for Stranded Vehicles
Specific Location	National highways and major roads in India; initial localized study context: Telangana/Hyderabad-region road travel.
Specific Issue	Vehicles can become stranded after fuel depletion, particularly when a suitable fuel source is not immediately accessible.
Specific Focus	Stranded vehicle user, nearby fuel source/retail outlet, emergency fuel delivery/support, location and request information, and coordination between parties.
Problem Area	A vehicle that runs out of fuel may be unable to continue its journey. The user may need to identify a nearby fuel source and coordinate assistance while remaining safely at the roadside.

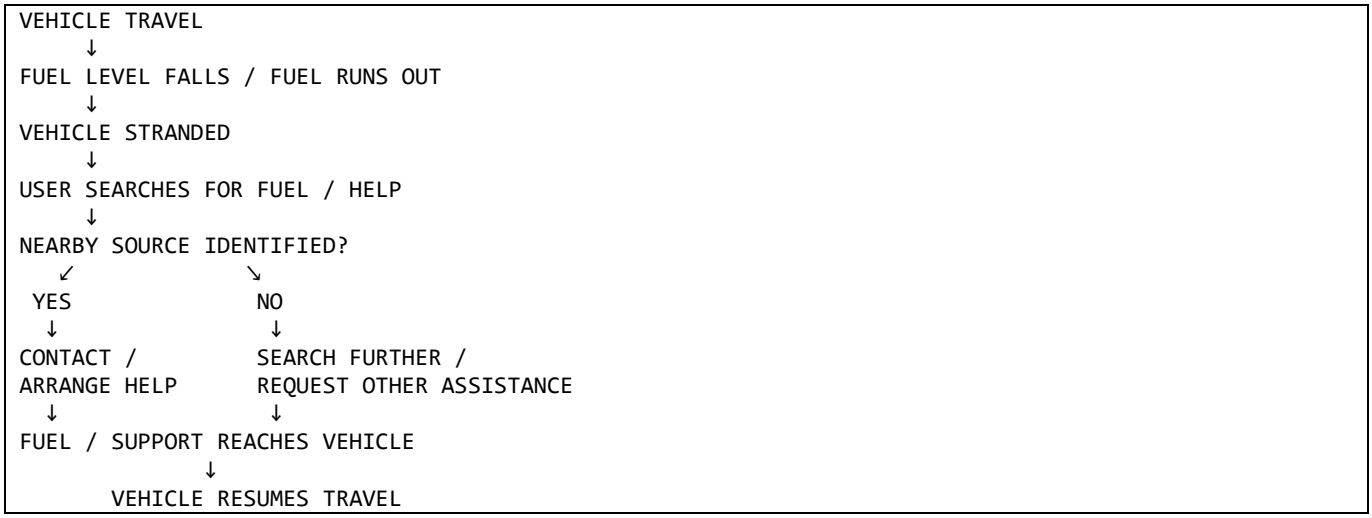
2. Study the System – 5W2H

5W2H	Filled response
WHAT – Symptoms / effects	Vehicle stops because usable fuel is insufficient; journey is interrupted; user must search for fuel or roadside assistance; waiting and coordination may increase; the stranded vehicle may remain at the roadside until assistance is obtained.
WHO – Affected	Stranded drivers/passengers; vehicle owners; families/travel companions; other road users when a stopped vehicle creates an obstruction or safety concern.
WHO – Contributing / involved stakeholders	Fuel-station/retail-outlet operators, fuel delivery/support personnel, roadside-assistance providers, highway authorities/road operators, emergency/support services, payment/service platforms.
WHEN	When a vehicle reaches an insufficient-fuel state. Risk can be higher during late-night/early-morning travel, long-distance travel, unfamiliar routes, or periods when nearby services are unavailable; these time patterns must be validated through field data.
WHERE	National highways and major roads, especially stretches where a suitable fuel source is not immediately accessible. Initial study can use selected Telangana/Hyderabad-region highway stretches.
HOW – observed process	Vehicle travels → fuel level becomes critically low → vehicle stops/strands → user searches for nearby fuel/help → identifies possible source →

	contacts/requests assistance → fuel/support is arranged → fuel reaches vehicle → vehicle resumes travel.
HOW MUCH – impact	Primary measurable impacts: waiting time, distance to nearest suitable fuel source, number of requests requiring assistance, assistance success rate, service cost, and duration of vehicle immobilization. Actual incident counts/costs should be collected through interviews, surveys, roadside-assistance records or other cited datasets.

3. As-Is Process / Lifecycle

As-Is lifecycle diagram



Note: This diagram describes the observed/problem-state process, not the final proposed solution.

4. Preliminary Problem Statement (<100 words)

On national highways and major roads, a vehicle may become stranded when its available fuel is insufficient to continue the journey and a suitable fuel source is not immediately accessible. The stranded user must locate a nearby source and coordinate assistance, which can create uncertainty, waiting, additional travel, and roadside-safety concerns. The problem involves the interaction of stranded users, fuel retail outlets/sources, assistance providers, road infrastructure, location information, communication, and payment/coordination processes. The project therefore studies the system around emergency fuel access rather than assuming a technology-first solution.

5. Problem Relationships / Causal Relationship

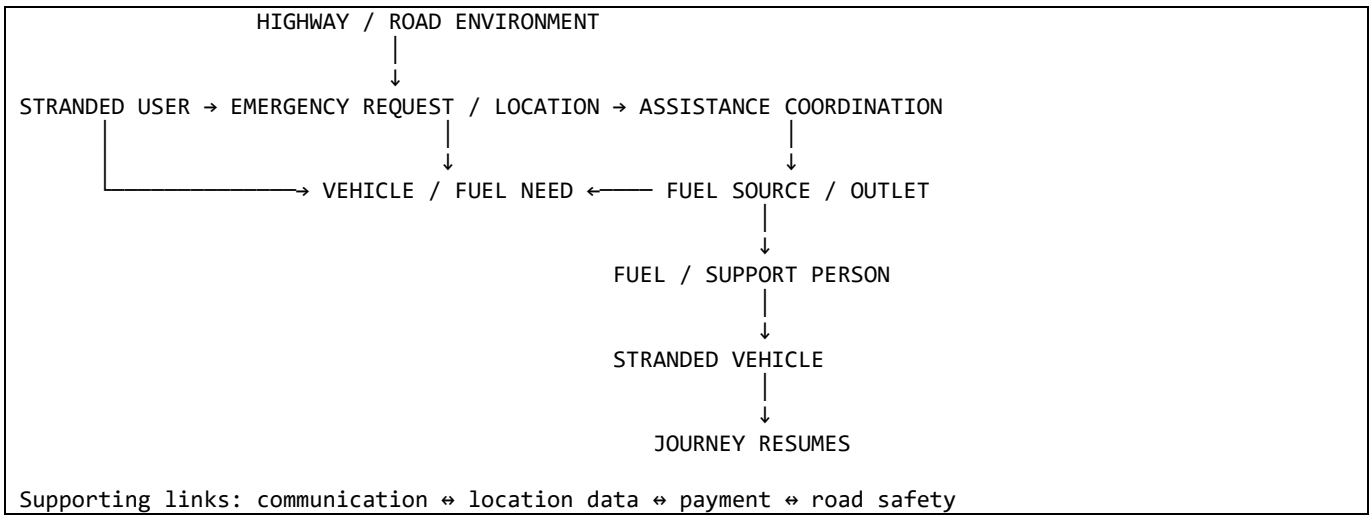
- Low available fuel → higher probability of vehicle becoming stranded.
- Vehicle stranded → need for fuel-source identification and assistance.
- Limited or uncertain access to a nearby source → longer search/coordination time.
- Longer waiting time → longer vehicle immobilization and potentially greater roadside exposure.
- Repeated difficult experiences can encourage drivers to seek informal/quick fixes rather than reliable long-term preparedness or assistance channels.

6. Define the System – Components

Component / role	Function
Stranded vehicle user	Reports the fuel emergency, provides location and vehicle/fuel information, receives assistance.
Vehicle	Physical object requiring fuel to resume travel.
Fuel retail outlet/source	Potential source of petrol/diesel or relevant fuel.
Fuel delivery/support person	Carries/coordinates fuel or assistance to the stranded vehicle, subject to applicable safety and legal requirements.
Location information	Identifies where the vehicle and possible sources are located.
Communication channel	Transfers request, status, directions and updates between stakeholders.
Request/dispatch process	Records the emergency and coordinates assistance.
Payment mechanism	Handles service/fuel payment where applicable.
Highway/road environment	Physical context in which the stranded vehicle and assistance movement occur.
Roadside safety/emergency services	Relevant support when the stopped vehicle creates a safety risk or when fuel assistance alone is insufficient.

7. System Map

System map – relationships between components

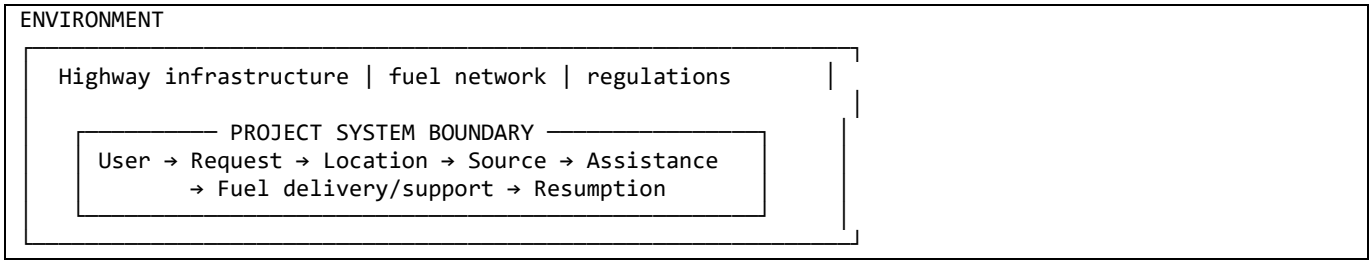


8. System Boundary / Scope

Inside the study boundary	Outside / environment
Stranded vehicle user and vehicle	Vehicle manufacturing/design and engine engineering.
Emergency fuel request and information flow	National fuel-policy decisions and petroleum production.
Nearby fuel-source identification	General road construction projects not directly related to access.
Coordination with fuel source/support provider	Unrelated vehicle breakdowns such as major

	mechanical failure, unless they affect fuel assistance.
Assistance dispatch and status updates	Long-term fuel-price policy and national energy planning.
Arrival of fuel/support and journey resumption	Full emergency medical response, except as a safety escalation.

System boundary



9. Primary Stakeholder Map

Stakeholder	Primary needs / expectations
1. Stranded vehicle user	Fast identification of feasible help; clear status; safe assistance; transparent service/fuel cost; reliable communication.
2. Fuel retail outlet / fuel source	Clear request details; confirmation of fuel availability; safe and compliant handling; predictable coordination/payment.
3. Fuel delivery / roadside-support provider	Accurate vehicle location; clear quantity/type of fuel required; safe access; efficient dispatch; payment/service confirmation.
Additional: Highway/road authority	Roadside safety, reduced obstruction, orderly incident handling, useful incident information.
Additional: Emergency/support services	Rapid escalation when the situation becomes a road-safety or emergency incident.

10. Quantitative Baseline Data

Use these cited infrastructure baselines in the PID. They establish the surrounding fuel/road system but do not by themselves prove how often vehicles run out of fuel. Incident frequency must be measured separately.

Data point	Value	Why relevant	Source
National Highway network in India	146,145 km (2024)	Shows the scale of the arterial highway network in which roadside fuel-access problems can occur.	MoRTH Annual Report 2023–24
Telangana petroleum retail outlets	4,189 as of 30 Sep 2024	Shows the size of the state fuel-retail network that users may depend on.	PPAC Telangana Oil & Gas Snapshot, 2024–25
Telangana CNG stations	205 as of 30 Sep 2024	Shows additional fuel/energy infrastructure in the state; useful for understanding fuel-type coverage.	PPAC Telangana Oil & Gas Snapshot, 2024–25

Telangana EV charging stations at retail outlets	996 as of 30 Sep 2024	Shows the changing roadside energy ecosystem and a possible future extension of roadside assistance beyond liquid fuels.	PPAC Telangana Oil & Gas Snapshot, 2024–25

Important evidence gap to collect locally: number of fuel-outage/stranding incidents, median distance to the nearest suitable fuel source, median waiting time, service cost, time-of-day distribution, and percentage of requests successfully fulfilled.

11. How Might We? (HMW)

1. How might we reduce the time and uncertainty involved in getting suitable fuel assistance to a vehicle stranded on a highway?

2. How might we connect a stranded vehicle with an appropriate nearby fuel source/support provider while keeping location, fuel type, quantity, safety and payment information clear?

3. How might we make emergency fuel assistance reliable across different highway conditions, service availability and time-of-day patterns without assuming a technology solution before validating the system need?

MODULE 2 – Systems Mapping & Root Cause

1. Iceberg Model – Emergency Fuel Assistance

Iceberg Model

EVENT (VISIBLE)

Vehicle runs out of fuel and becomes stranded

↓

PATTERNS

- Repeated fuel-access/search problems on long road trips
- Longer search/waiting when nearby suitable sources are unclear
- Greater dependence on communication and location accuracy

↓

STRUCTURES

- Spatial distribution of fuel outlets
- Road network and access constraints
- Information/communication gaps
- Availability of assistance providers
- Fuel-type/quantity matching constraints
- Safety, compliance and payment processes

↓

MENTAL MODELS

- “A fuel station will always be nearby.”
- “The driver can solve the issue by searching manually.”
- “Location and fuel details can be communicated later.”
- “Emergency assistance is mainly a one-off transaction.”

2. Visible Events / Issues

- Vehicle stops because fuel is insufficient.
- Driver/passengers search for a fuel source or assistance.
- Potential fuel sources may be too far, unsuitable for the vehicle/fuel type, closed/unavailable, or difficult to access.
- Assistance requires exchange of location, fuel requirement and contact information.
- Vehicle remains immobilized until fuel/support reaches it.
- Journey is delayed and roadside exposure continues during the waiting period.

3. Events → Patterns

Observed events	Recurring pattern to investigate
Vehicle becomes stranded after fuel depletion	Fuel-related stranding can recur across long-distance/highway travel; frequency must be validated with field data.
User searches for nearby fuel/help	Manual search and coordination may create variable response times.
Assistance depends on source availability and access	Response depends on spatial and temporal availability of fuel/support.
Vehicle remains stopped while help is arranged	Waiting/immobilization is a key measurable outcome.

4. Stakeholder Groups from Module 1

Stakeholder	Needs	Concerns	Non-negotiables
Stranded user	Fast, clear, safe assistance; correct fuel; transparent cost.	Waiting, wrong location, wrong fuel, unreliable provider.	Safety, correct fuel, clear communication.
Fuel source/outlet	Accurate request and fuel requirement; safe transaction.	Unclear demand, payment uncertainty, unsafe roadside handling.	Safety/compliance and correct transaction.
Assistance provider	Accurate location; clear fuel quantity/type; feasible route.	Wrong location, inaccessible vehicle, unclear request.	Safety, feasible access, correct request information.

5. Stakeholder Process & Lifecycle

Stakeholder	Process sequence	Lifecycle
Stranded user	Travel → fuel becomes insufficient → vehicle stops → assess situation → share location/fuel need → request/coordinate help → wait → receive fuel/support → resume journey.	One incident from stranding until journey resumes; repeats whenever similar incidents occur.
Fuel source/outlet	Receive request → confirm fuel/source availability → confirm quantity/type → prepare/hand over fuel/support → payment/record → close request.	Starts at request and ends after the fuel/service transaction is completed.
Assistance provider	Receive dispatch → accept/confirm → obtain location and fuel details → travel to vehicle → verify request → provide assistance → close job.	Starts at dispatch and ends after assistance is delivered.

6. System Structure

Process step	System structures enabling it
Vehicle travels	Road network, vehicle, fuel tank, driver behavior, trip planning.
Fuel becomes low	Fuel consumption, tank capacity, distance between accessible sources, driver monitoring habits.
Vehicle becomes stranded	Insufficient remaining fuel, lack of immediately accessible source, road/location context.
Request is created	Communication channel, user information, location information, request process.
Source/support is selected	Fuel availability, distance, access route, provider availability, fuel type/quantity.
Assistance travels	Road infrastructure, provider vehicle/person, traffic, distance, access restrictions.
Fuel/support delivered	Safe handling, correct fuel, transaction/payment, confirmation.
Journey resumes	Adequate fuel/support, vehicle operational condition, safe road access.

7. Mental Models

Stakeholder	Likely belief / expectation
Stranded user	Expects that a nearby fuel source or assistance option should be findable quickly; values certainty and safety.
Fuel source/outlet	Expects requests to contain enough information to serve the user correctly and safely.
Assistance provider	Expects accurate location and fuel details before dispatch; values feasible routes and efficient jobs.
Road authority	Expects stranded vehicles to be managed without creating additional traffic or safety hazards.

8. Key Bottlenecks

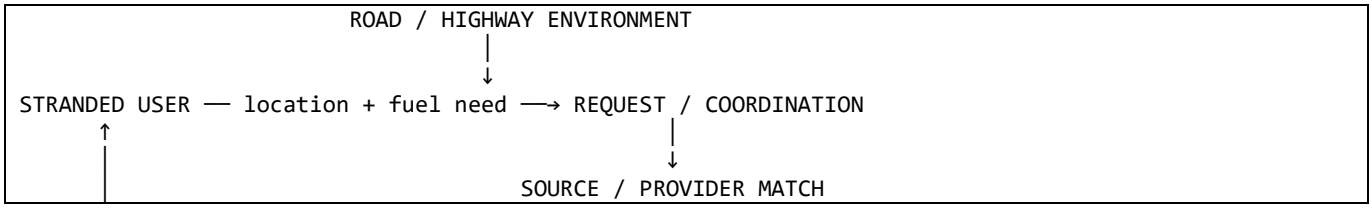
Type	Bottleneck identified	Possible measurement
Data bottleneck	Uncertainty about exact vehicle location, fuel type/quantity and source availability.	% requests with complete information; location error; source confirmation time.
Process bottleneck	Multiple manual steps may be needed to find, contact and coordinate a source/provider.	Request-to-dispatch time; dispatch-to-arrival time.
Human bottleneck	Driver may be stressed, unfamiliar with location, or unable to coordinate while managing roadside safety.	Time spent searching/contacting; failed contacts; user-reported difficulty.
Infrastructure bottleneck	Fuel sources and assistance providers are spatially uneven and may have time-dependent availability.	Distance to suitable source; open/available status; coverage by road segment.

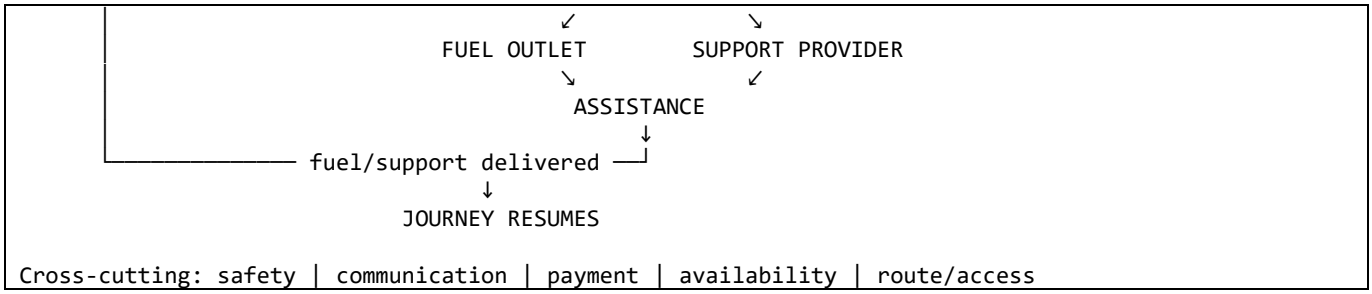
9. Review of HMW Against Bottlenecks

HMW	Bottlenecks addressed
HMW 1 – reduce time/uncertainty	Process + human bottlenecks: simplify the sequence from emergency request to confirmed assistance.
HMW 2 – connect user to appropriate source/provider	Data + process bottlenecks: match location, fuel type, quantity, availability and route.
HMW 3 – make assistance reliable across conditions	Infrastructure + process bottlenecks: account for time, source availability and changing road conditions.

10. Revised System Map

Completed Module 2 system map




FUEL OUTLET
SUPPORT PROVIDER
ASSISTANCE
fuel/support delivered
JOURNEY RESUMES
Cross-cutting: safety | communication | payment | availability | route/access

MODULE 3 – Causal Loop Diagrams & Leverage

1. CLD Variables

Variable	Direction / meaning
Fuel availability near route	Higher value = more accessible suitable fuel sources.
Ease of finding suitable source	Higher value = easier/faster source identification.
Search/coordination time	Higher value = more time spent before assistance is confirmed.
Assistance response time	Higher value = longer time before fuel/support reaches vehicle.
Vehicle immobilization time	Higher value = longer period vehicle remains stopped.
Roadside exposure/risk	Higher value = greater exposure to traffic/environmental risk.
Accurate request information	Higher value = better location, fuel type, quantity and contact information.
Provider/source availability	Higher value = more feasible assistance options.
Successful assistance rate	Higher value = more requests completed successfully.

2. Reinforcing Loop (R1) – Delay and Roadside Exposure

Reinforcing Loop R1

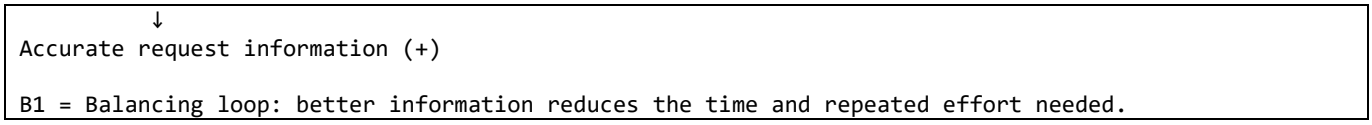
```
Search / coordination time (+)
          ↓
Assistance response time (+)
          ↓
Vehicle immobilization time (+)
          ↓
Roadside exposure / uncertainty (+)
          ↓
Difficulty coordinating safely (+)
          ↓
Search / coordination time (+)

R1 = Reinforcing loop: delays can reinforce further coordination difficulty.
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3. Balancing Loop (B1) – Faster Matching

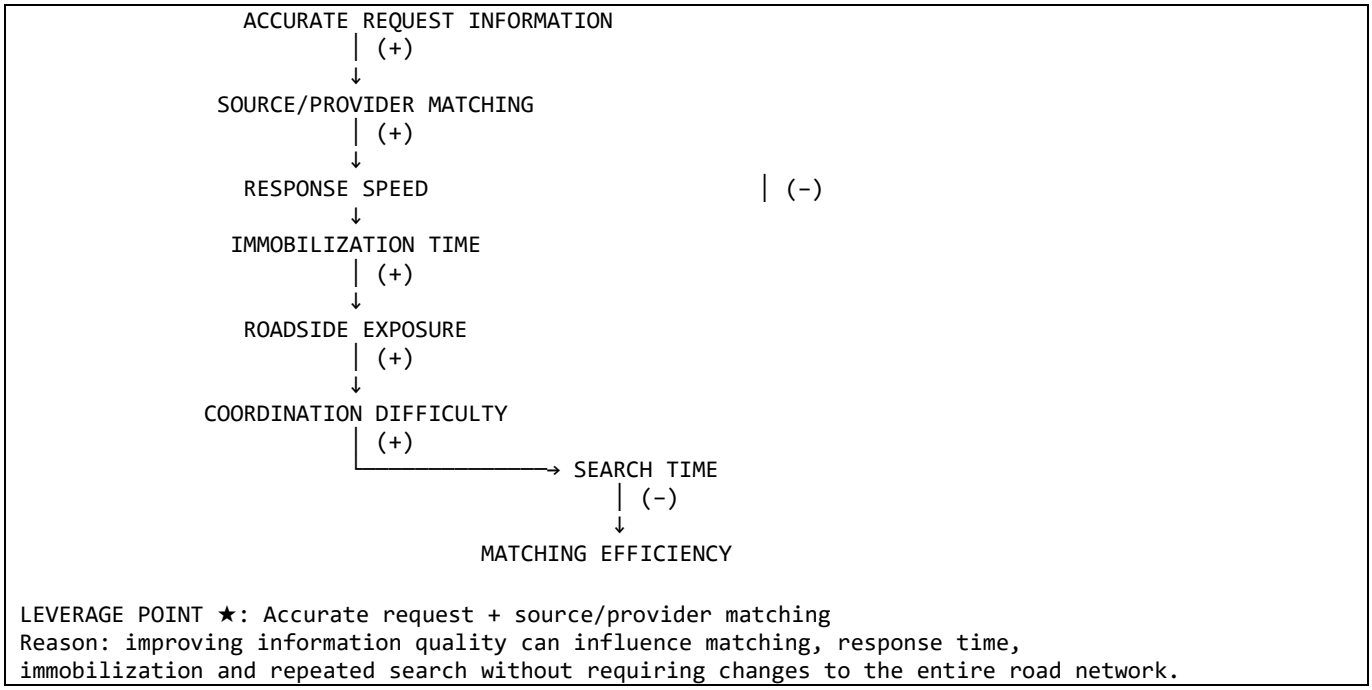
Balancing Loop B1

```
Accurate request information (+)
          ↓
Ease of finding suitable source (+)
          ↓
Provider/source confirmation speed (+)
          ↓
Assistance response time (–)
          ↓
Vehicle immobilization time (–)
          ↓
Need for repeated searching (–)
          ↓
Search / coordination time (–)
```



4. Combined Causal Loop Diagram & Leverage Point

Combined CLD – ★ marks the proposed leverage point



5. Leverage Point

Chosen leverage point: improve the quality and timeliness of information used to match a stranded vehicle with an appropriate nearby fuel source/support provider. The key information includes vehicle location, fuel type, required quantity, contact details, source/provider availability, route/access constraints and request status.

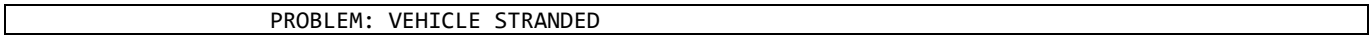
Why this is a leverage point: the workbook defines a leverage point as a place where a relatively small change can create a large shift in system behaviour. Better information can reduce repeated searching, improve matching and shorten coordination/response time.

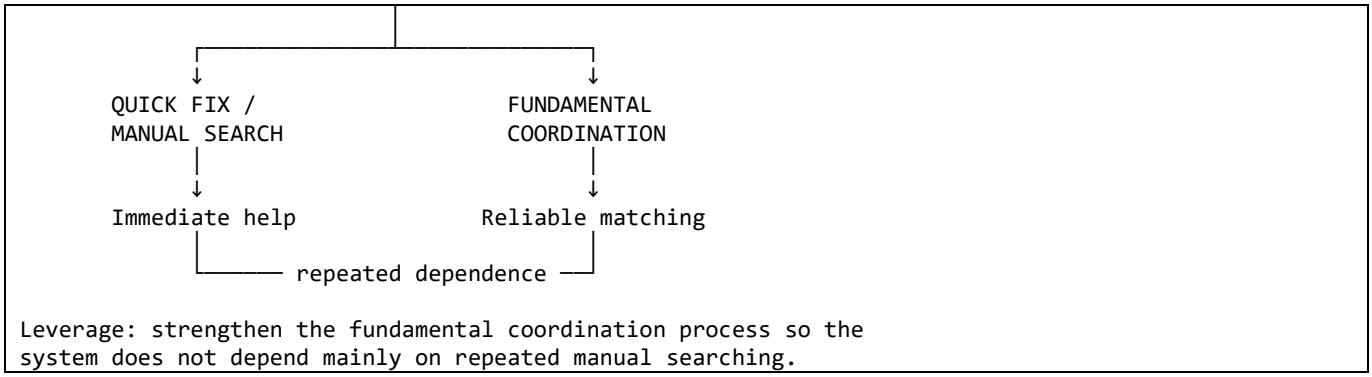
6. System Archetype

Best-fit archetype: Shifting the Burden.

Reason: the immediate response to fuel stranding is often a quick/manual search for a fuel source or person who can help. That short-term response may solve the immediate incident, but it does not by itself create a reliable, repeatable way of connecting the stranded user to appropriate assistance. The long-term improvement is to strengthen the underlying coordination and information structure so that repeated manual searching is reduced.

Shifting the Burden interpretation





7. Module 3 Questions – Final Answers

Question	Answer
Which archetype best describes the system problem?	Shifting the Burden.
Why does it fit?	The immediate/manual search can address a single stranded-vehicle incident, while the deeper coordination and information structure determines whether assistance can be obtained reliably and repeatedly.
Reinforcing loop identified?	R1 – longer search/coordination → longer response → longer immobilization/roadside exposure → greater coordination difficulty → longer search/coordination.
Balancing loop identified?	B1 – accurate information → easier matching → faster confirmation → lower response/immobilization → less repeated searching.
Leverage point?	Accurate emergency-request information combined with appropriate source/provider matching.

8. Evidence and Validation Plan

- Conduct interviews with stranded drivers/vehicle owners to measure search time, waiting time, common difficulties and preferred assistance channels.
- Survey selected highway users on how they would obtain fuel assistance if stranded.
- Record the distance to the nearest suitable fuel source for selected road segments.
- Record source/provider availability by time of day where data can be obtained.
- Measure request-to-confirmation and confirmation-to-arrival times in a pilot.
- Do not claim that the proposed system reduces incidents until before/after data are collected.

9. Source Notes for PID

The workbook requires claims to be evidence-backed and asks for official/recognised sources. The following government sources support the baseline infrastructure figures used above.

Source	Use in workbook
Ministry of Road Transport & Highways (MoRTH), Annual Report 2023–24	National Highway network baseline: 146,145 km in 2024.
Petroleum Planning & Analysis Cell (PPAC), Telangana Oil & Gas Snapshot, Volume 2024–25, data as of 30	Telangana retail outlets: 4,189; CNG stations: 205; EV charging stations at retail outlets: 996.

Sep 2024	

Web references checked: MoRTH Annual Report 2023–24 and PPAC Telangana Oil & Gas Snapshot 2024–25. Use the official source documents/links in the final college submission bibliography.

10. Important Submission Note

The workbook explicitly says the PID is an As-Is representation and should not be mixed with the 'should-be' state. Therefore, the completed Modules 1–3 above deliberately separate observed/current-state statements from the proposed leverage point and future system intervention. Replace the marked evidence gaps with your team's actual interview/survey/road-observation data before final submission.